



General Chemistry I

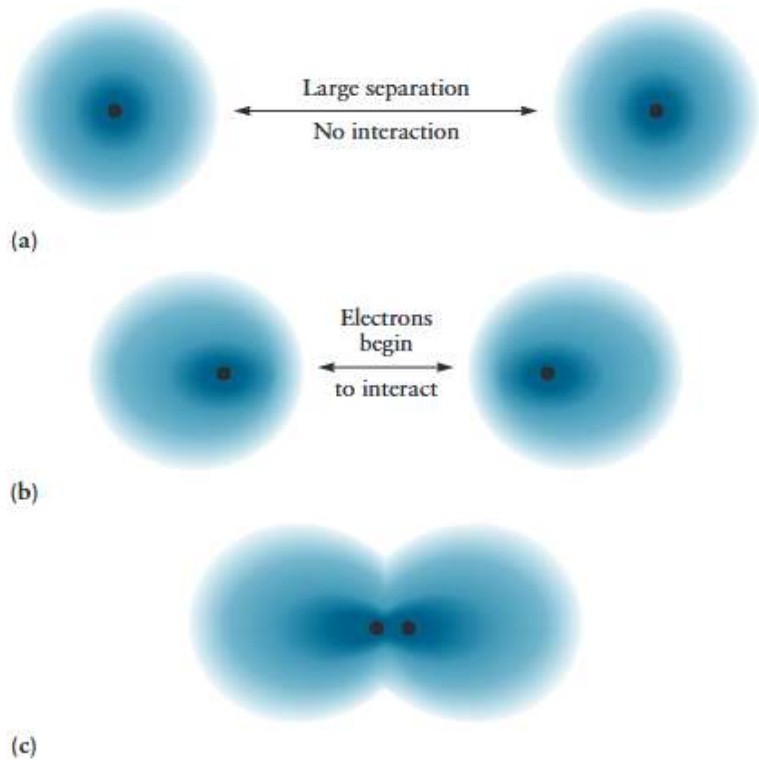
Recitation 8

Siyuan Chen
Nov. 17th

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
															Pnictogens	Chalcogens	Halogens			
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026		
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122													5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305													13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798		
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29		
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)		
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)		
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																				
			6 7	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97		
				89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)		

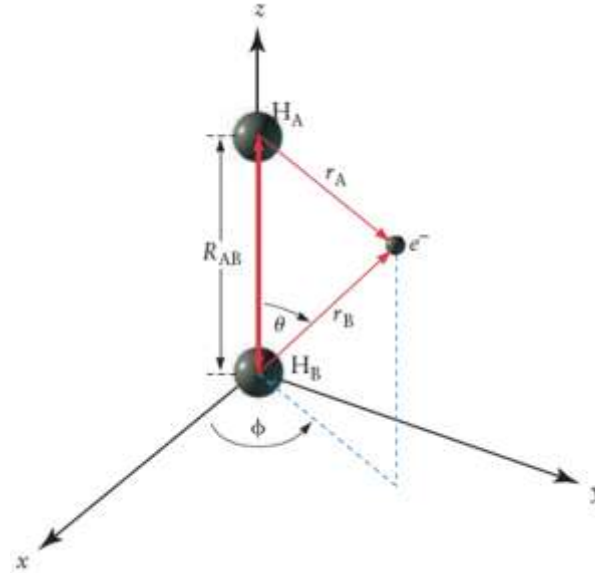
MO and molecular structure

Mechanism of covalent Bond formation

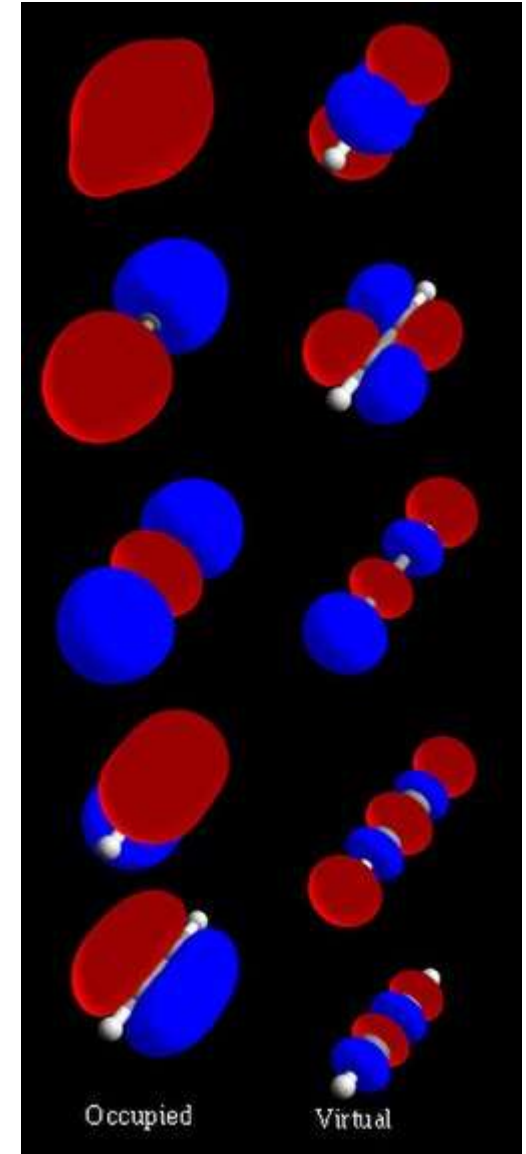


At very large distances the electron densities of the two H atoms are independent. As the distance becomes sufficiently small, the two densities interpenetrate to form a new pattern of electron density characteristic of the H_2 molecule

MO: Regions in a molecule where an electron occupying that orbital is likely to be found



In an isolated atom, the orbital electrons' **location** is determined by functions called **atomic orbitals**. When multiple atoms combine chemically into a molecule by forming a valence chemical bond, the electrons' **locations** are determined by the molecule **as a whole**, so the atomic orbitals combine to form molecular orbitals. The electrons from the constituent atoms occupy the molecular orbitals.

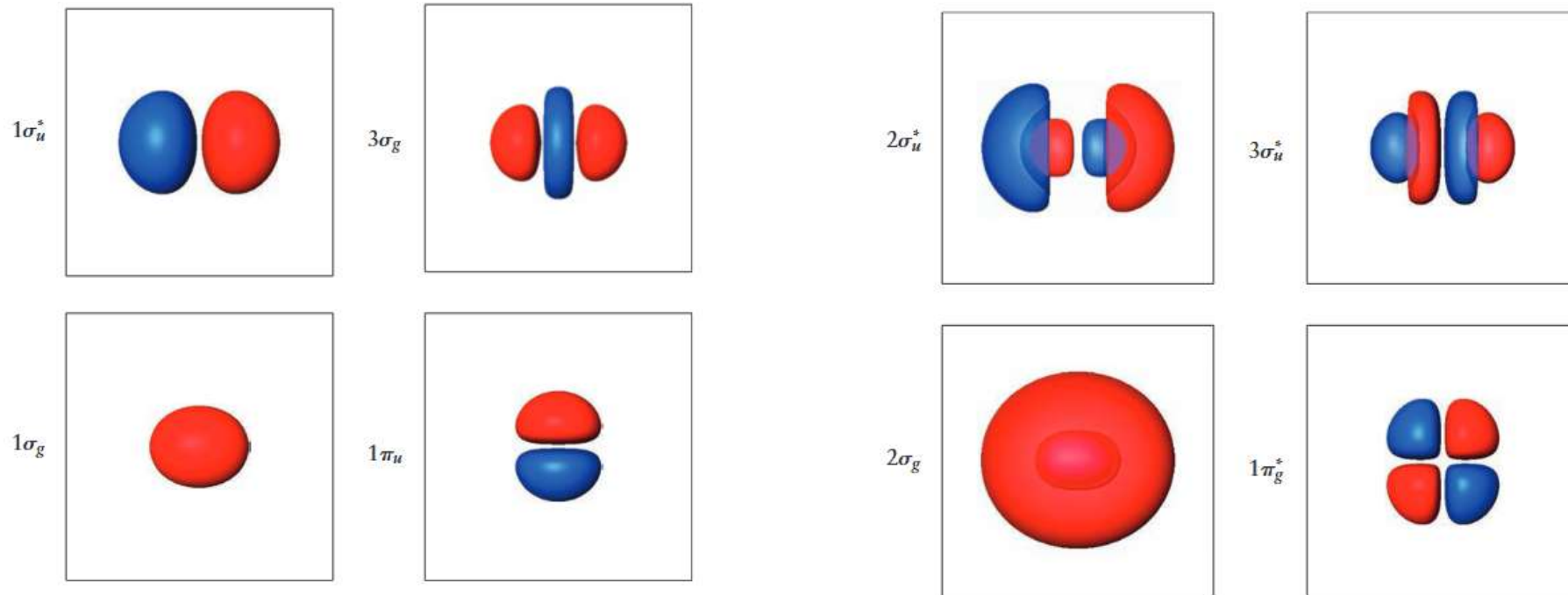


MO and molecular structure

Molecular orbitals are, in general, delocalized throughout the entire molecule. Moreover, if the molecule has symmetry elements, its nondegenerate molecular orbitals are either symmetric or antisymmetric with respect to any of these symmetries.

eg: H_2O

The symmetry properties of molecular orbitals means that **delocalization** is an inherent feature of molecular orbital theory and makes it fundamentally different from (and complementary to) valence bond theory, in which bonds are viewed as **localized electron pairs**, with allowance for resonance to account for delocalization.



The sign of the wave function is the same at the point (x, y, z) and at the point $(-x, -y, -z)$, it is called symmetric and labeled g

gerade and ungerade(中心对称/中心反对称)

MO and molecular structure

MO theory

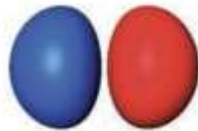
Molecular orbitals arise from allowed interactions between atomic orbitals (Atomic orbital overlap)

How to form efficient overlap:

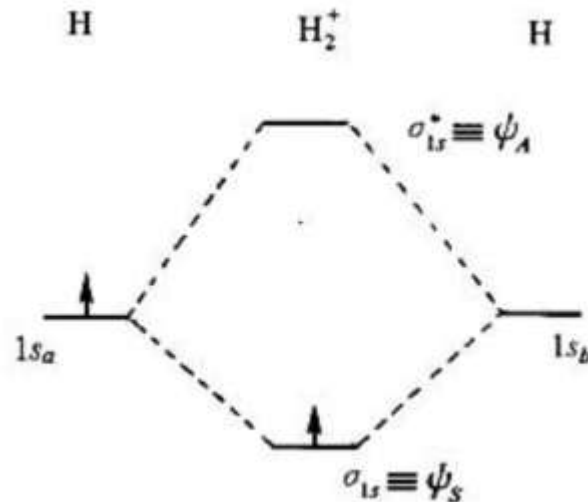
- symmetries are compatible with each other.
- atomic orbitals are close in energy.
- the number of molecular orbitals formed must be equal to the number of atomic orbitals in the atoms being combined to form the molecule.



$1\sigma_g$



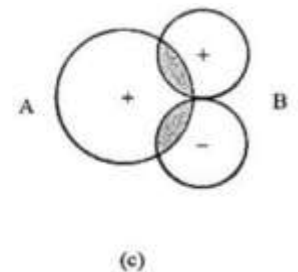
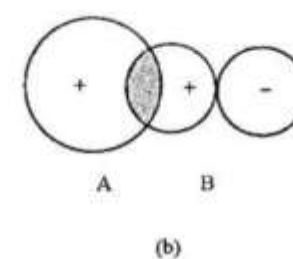
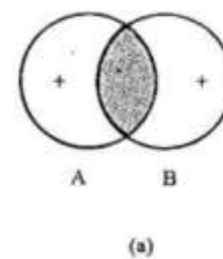
$1\sigma_u^*$



Linear Combination of Atomic Orbitals

$$1\sigma_g \approx \sigma_{g1s} = C_g[\varphi_{1s}^A + \varphi_{1s}^B]$$

$$1\sigma_u^* \approx \sigma_{u1s}^* = C_u[\varphi_{1s}^A - \varphi_{1s}^B]$$



MO and molecular structure

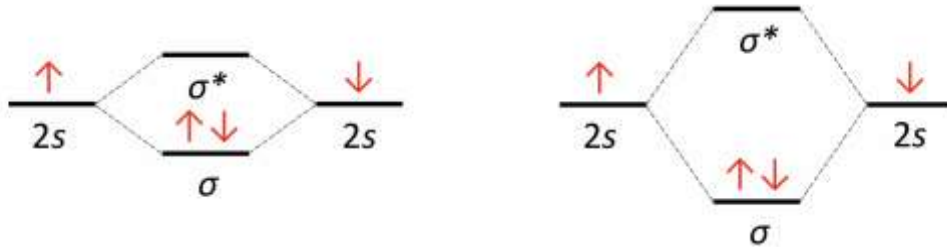
MO theory

Molecular orbitals arise from allowed interactions between atomic orbitals (Atomic orbital overlap)

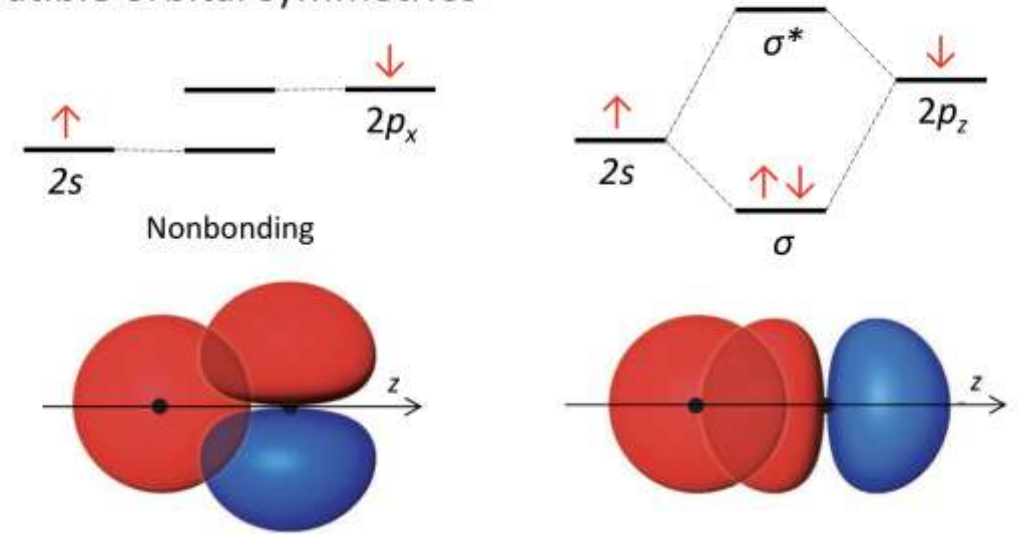
How to form efficient overlap:

- symmetries are compatible with each other.
- atomic orbitals are close in energy.
- the number of molecular orbitals formed must be equal to the number of atomic orbitals in the atoms being combined to form the molecule.

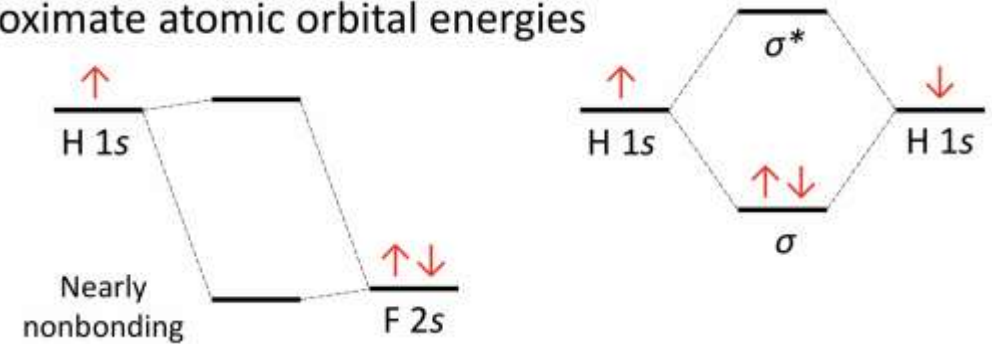
① Maximum orbital overlap



② Compatible orbital symmetries

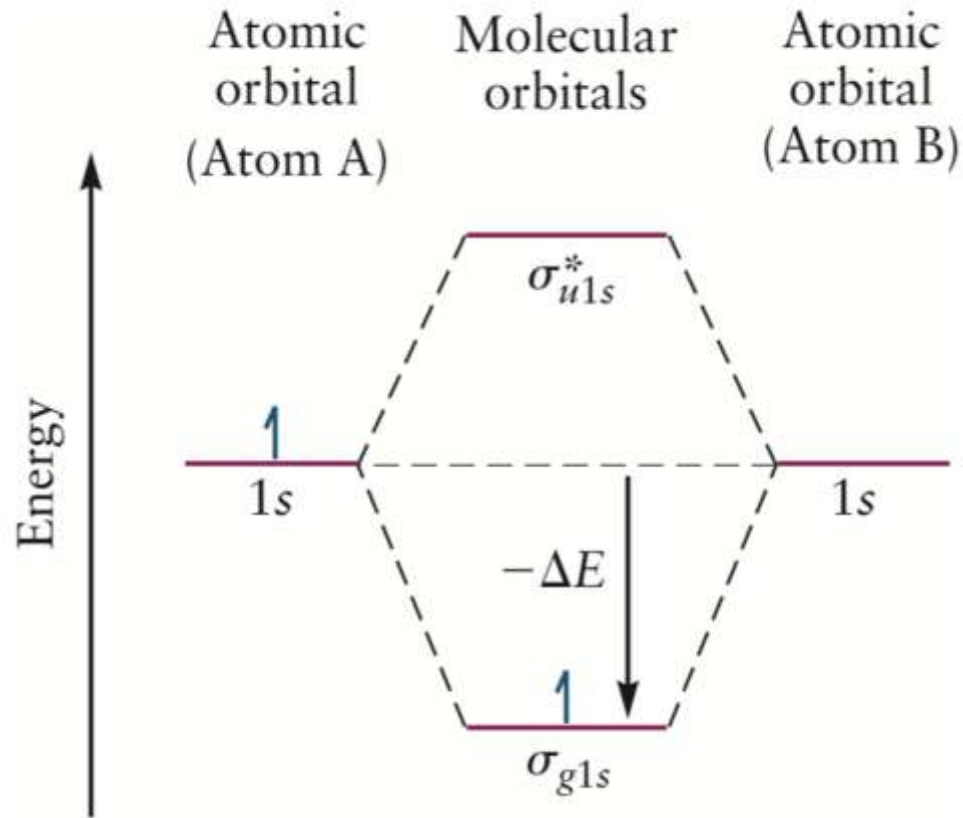


③ Approximate atomic orbital energies



MO and molecular structure

How to fill the orbitals with electrons?



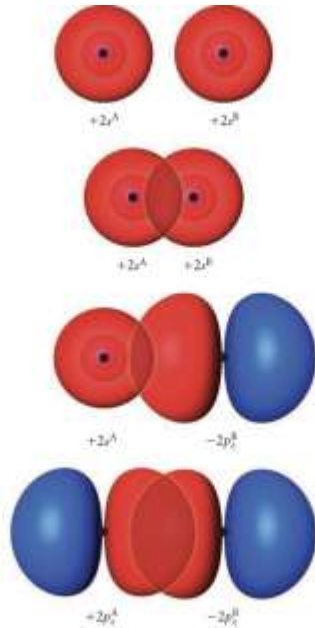
- Construct all molecular orbitals.
- Half bonding orbitals and half anti-bonding orbitals (what will happen if orbital number is odd?)
- Identify all valence electrons of the molecule.
- Populate the electrons in order of increasing energy (taking care to adhere to Hund's rule).

MO and molecular structure

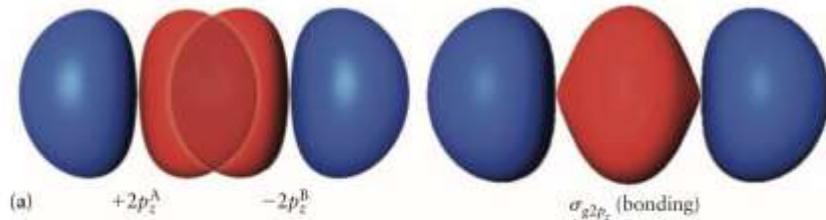
2s AOs of the two atoms, giving a σ_{g2s} bonding orbital and a σ_{u2s}^* antibonding orbital

$$\sigma_{g2s} = C_g[2s^A + 2s^B]$$

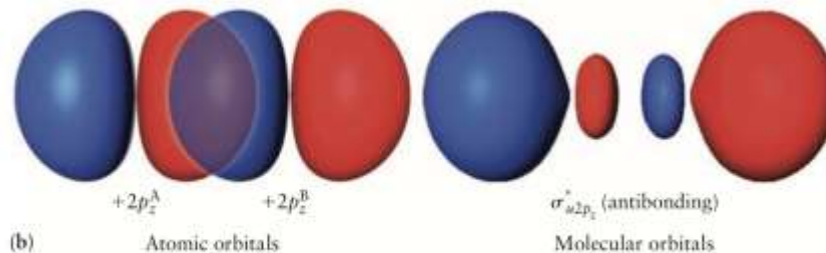
$$\sigma_{u2s}^* = C_u[2s^A - 2s^B]$$



Because σ bonds feature greater overlap than π bonds, σ bonding and σ^* antibonding orbitals feature greater energy splitting (separation) than π and π^* orbitals.

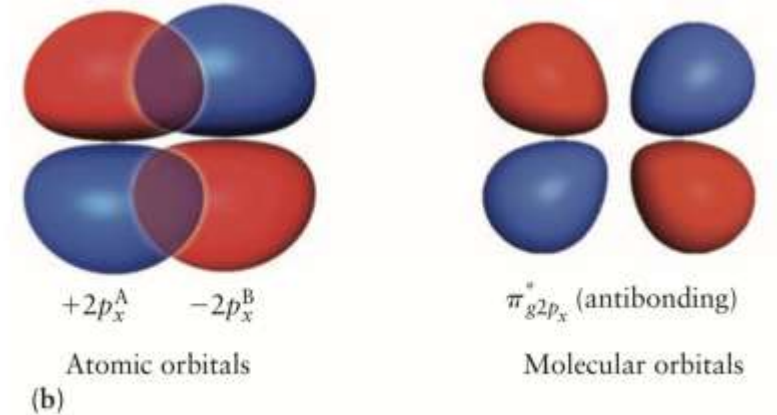
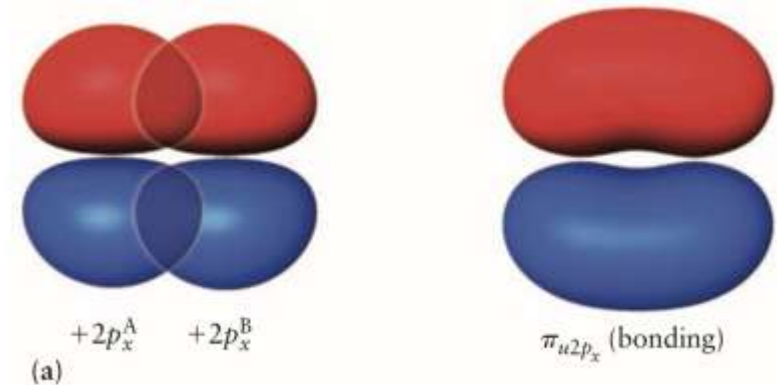


2p orbitals form different MOs depending on whether they are parallel or perpendicular to the internuclear (bond) axis



$$\sigma_{g2p_z} = C_g[2p_z^A - 2p_z^B]$$

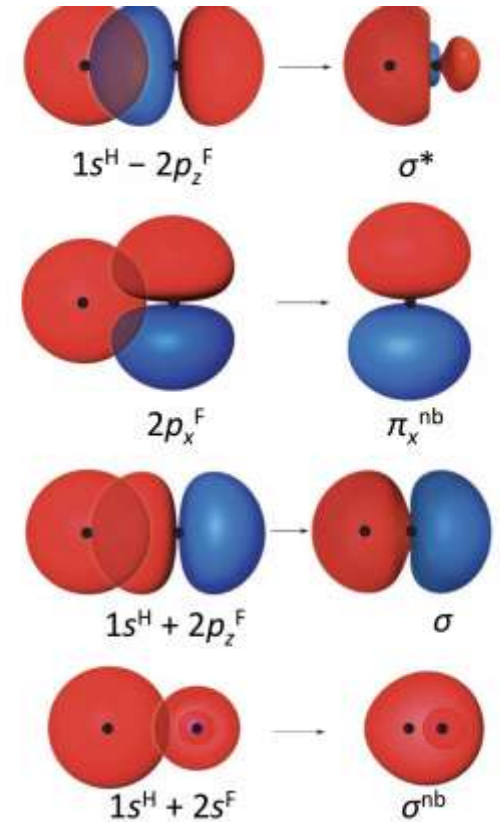
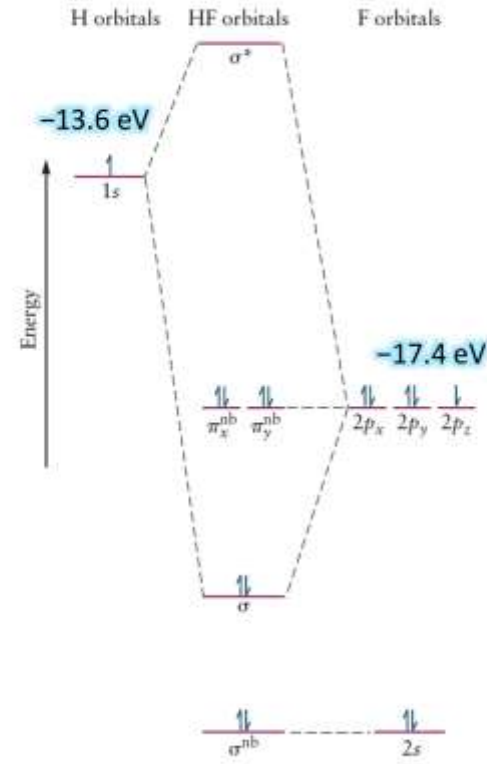
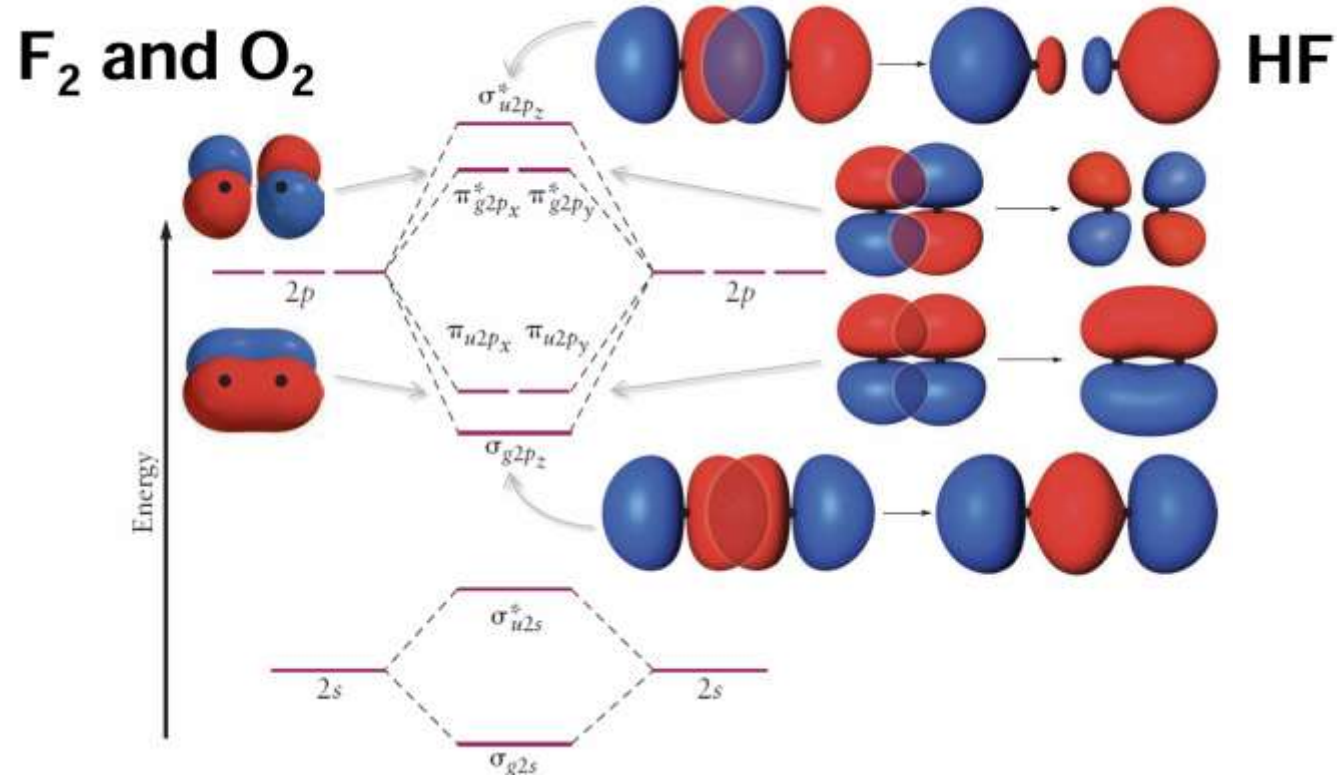
$$\sigma_{u2p_z}^* = C_u[2p_z^A + 2p_z^B]$$



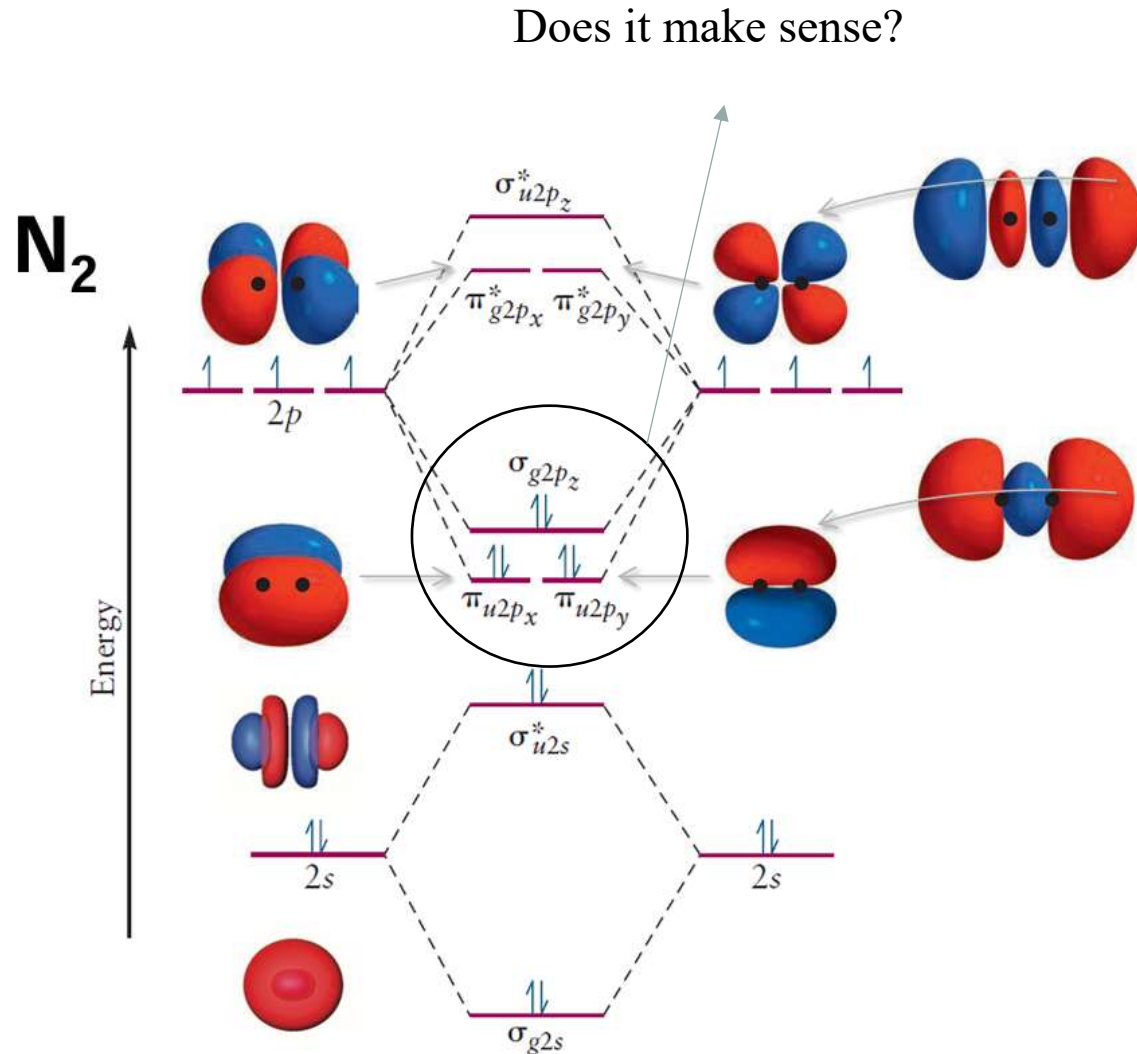
π_u or π_g ?

MO and molecular structure

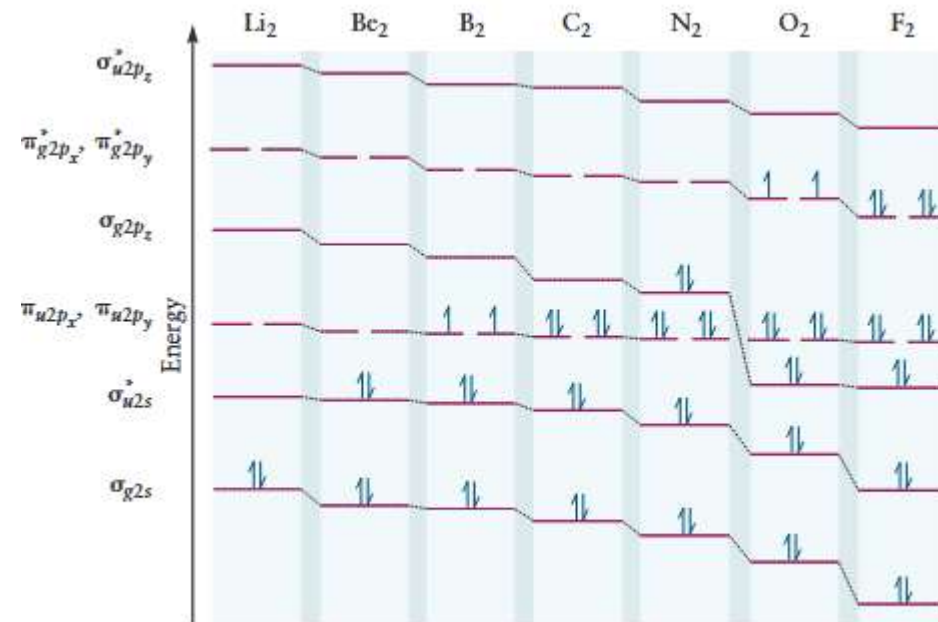
Try to sketch a molecular orbital



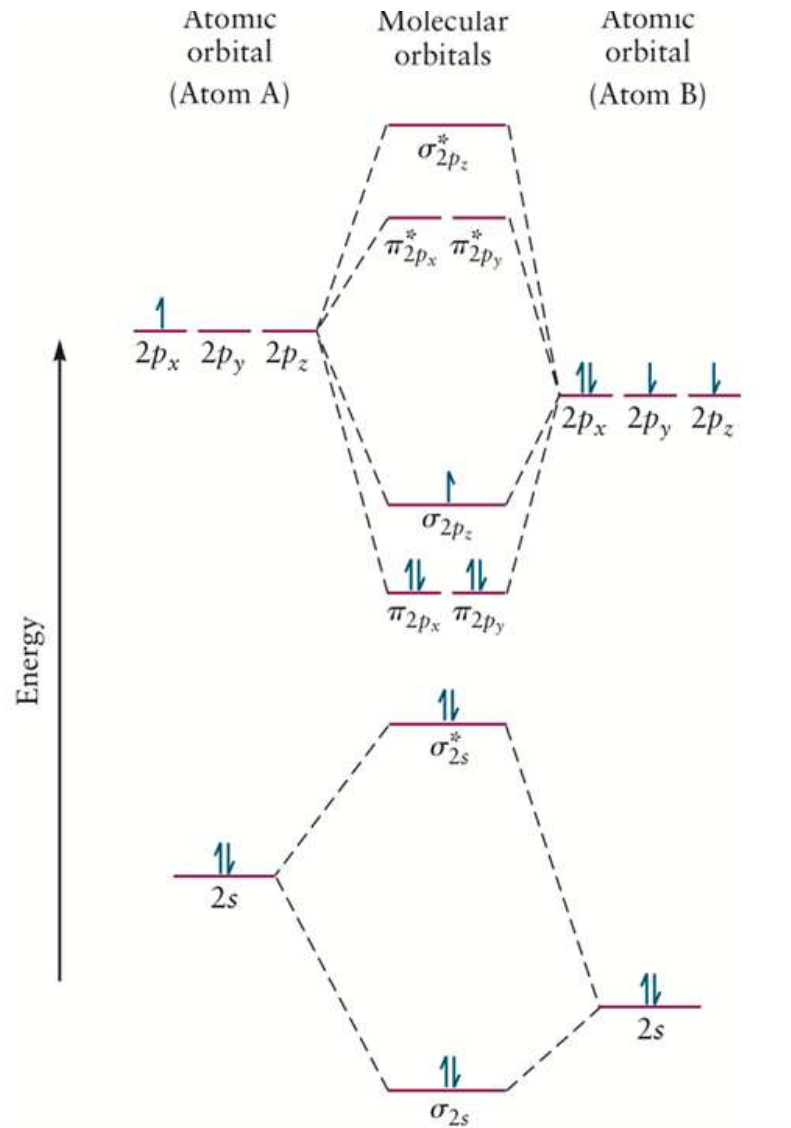
MO and molecular structure



The phenomenon of s-p mixing occurs when molecular orbitals of the **same symmetry** formed from the **combination of 2s and 2p atomic orbitals** are close enough in energy to further interact, which can lead to a change in the expected order of orbital energies.



MO and molecular structure

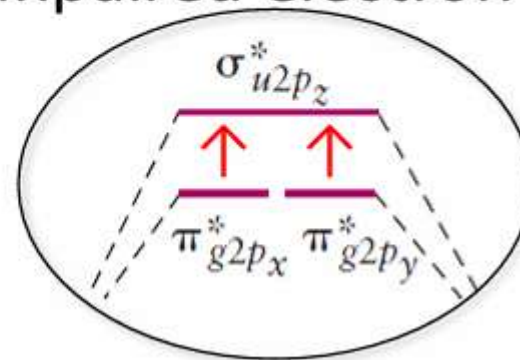


What can we learn from MO diagrams?

- Understand the bonding situation
- Deduce magnetic properties
- How molecules can change with ionization
- Bond order

$$\text{bond order} = \frac{1}{2} (\text{number of electrons in bonding molecular orbitals} - \text{number of electrons in antibonding molecular orbitals})$$

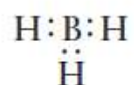
O₂ is paramagnetic due to its two unpaired electron



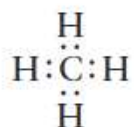
MO and VB theory



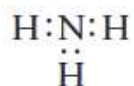
Beryllium hydride, BeH_2 , has four valence electrons, two from Be and one each from the two H atoms, all of which appear in the Lewis diagram. In VSEPR theory, the steric number is 2, so the molecule is predicted to be linear, and this prediction is verified by experiment. The electron configuration of the central atom is Be: $(1s)^2(2s)^2$. There are no unpaired electrons to overlap with H(1s) orbitals, so the VB model fails to predict the formation of BeH_2 .



Boron hydride, BH_3 , has six valence electrons corresponding to steric number 3 and a trigonal planar structure. With the electron configuration B: $(1s)^2(2s)^2(2p)^1$ on the central atom, the VB model cannot account for the formation of BH_3 and, in fact, predicts that BH is the stable molecule, which does not agree with experiment.



Methane, CH_4 , has steric number 4, and VSEPR predicts a tetrahedral structure, which is confirmed by experiment. Starting with the electron configuration C: $(1s)^2(2s)^2(2p)^2$, the VB model cannot account for the formation of CH_4 and predicts that CH_2 would be the stable hydride, which is again contrary to the experimental results.

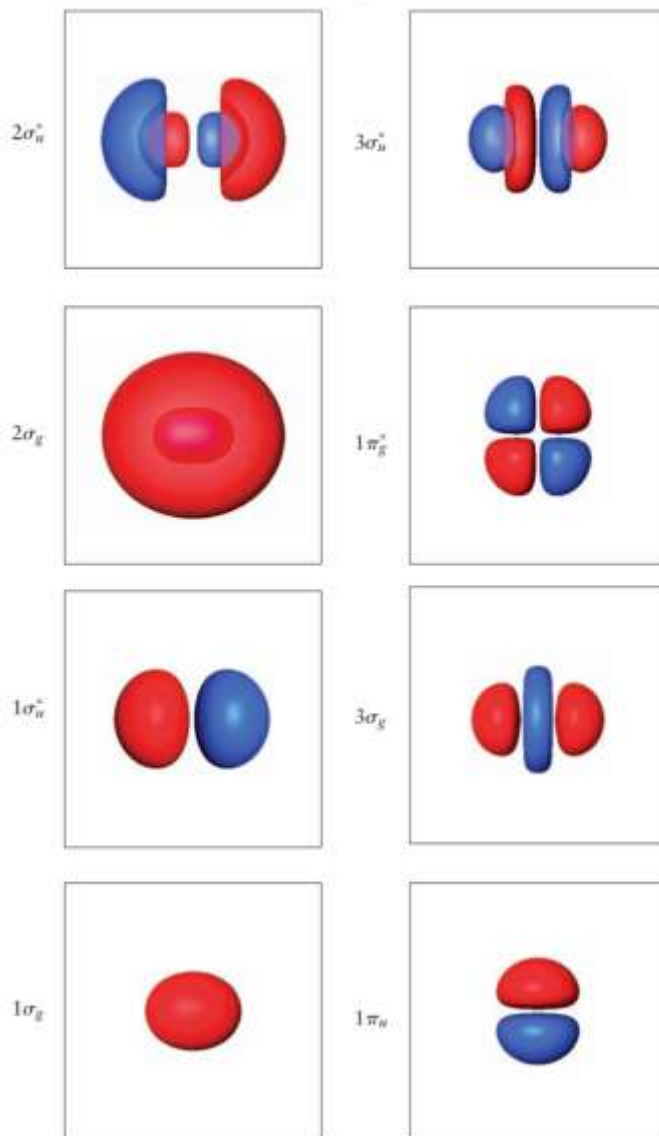


Ammonia, NH_3 , has steric number 4 with three shared pairs and one unshared pair on the N atom. VSEPR predicts a trigonal pyramid structure, as a subcase of tetrahedral structure, with angles slightly less than 109.5° due to repulsion between the unshared pair and the three bonding pairs. Experiment verifies this structure with angles of 107° . The electron configuration N: $(1s)^2(2s)^2(2p_x)^1(2p_y)^1(2p_z)^1$ would permit the formation of three σ bonds by overlap of H(1s) orbitals with each of the 2p orbitals on N. Because these 2p orbitals are all mutually perpendicular, VB predicts a trigonal pyramid but one with bond angles of 90° .

Homework4

- Determine the number of nodes along the internuclear axis and the number of nodal planes for each of the π molecular orbitals for H_2^+ shown in Figure 6.5.

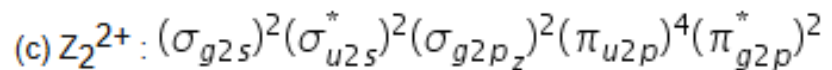
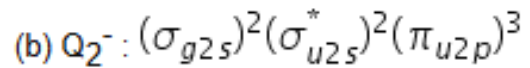
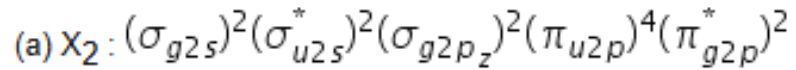
FIGURE 6.5 Wave functions for the first eight energy levels of the H_2^+ molecular ion, calculated exactly by quantum mechanics. The two nuclei lie along the z-axis, which is in the plane of the paper. Regions of positive and negative values of the wave function are shown in red and blue, respectively. The labels for each orbital are explained in the text. The images are isosurfaces corresponding to contours at ± 0.1 of the maximum value. The $1\sigma_g$ is the orbital the electron occupies when H_2^+ is in its ground electronic state, and the energies increase in the order $1\sigma_g < 1\sigma_u^* < 2\sigma_g < 2\sigma_u^* < 1\pi_u < 3\sigma_g < 1\pi_g^* < 3\sigma_u^*$. (Courtesy of Dr. Helen Hatal and Professor William A. Goddard III, California Institute of Technology; and Dr. Kelly E. Gubbler, University of Texas at Austin.)



- Predict the valence electron configuration and the total bond order for the molecule I_2 . Will I_2 be paramagnetic or diamagnetic?
- The compound nitrogen oxide (NO) forms when the nitrogen and oxygen in air are heated. Predict whether the nitrosyl ion (NO^+) will have a shorter or a longer bond than the NO molecule. Will NO^+ be paramagnetic like NO or diamagnetic?
- The photoelectron spectrum of CO has four major peaks with ionization energies of 14.5, 17.2, 20.1, and 38.3 eV. Assign these peaks of molecular orbitals of CO, and prepare a quantitative energy level correlation diagram for CO. The ionization energy of carbon atoms is 11.26 eV, and the ionization energy of oxygen atoms is 13.62 eV.
- Calcium carbide (CaC_2) is an intermediate in the manufacturing of acetylene (C_2H_2). It is the calcium salt of the carbide (also called acetylide) ion (C_2^{2-}). What is the electron configuration of this molecular ion? What is its bond order?

Quiz

1. For each of the following valence electron configurations of a homonuclear diatomic molecule or molecular ion, identify the element X, Q, or Z and determine the total bond order.



2. The molecular ion HeH^+ has an equilibrium bond length of 0.774 Å. Draw an electron correlation diagram for this ion, indicating the occupied MOs. Is HeH^+ paramagnetic? When HeH^+ dissociates, is a lower energy state reached by forming $\text{He} + \text{H}^+$ or $\text{He}^+ + \text{H}$?